SACG: Spatially Adaptive Consistency Gating for Refinement of Frozen Prompt-Based Metric Depth Estimates

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Working draft — October 2026

Abstract

Prompt-conditioned metric depth estimators combine image priors with low-resolution range measurements but can retain local depth errors. We study SACG, a lightweight refinement module trained while Prompt Depth Anything (PromptDA) remains entirely frozen. SACG uses RGB–depth gradient agreement and a learned spatial field derived from the validity of LiDAR samples to predict a gate, which modulates a depth residual estimated from RGB, coarse depth and frozen DPT features. Our selected variant is trained with valid-pixel L1 and a delayed gate penalty, without the boundary-weighted term of an earlier version. On 5,599 ARKitScenes validation frames (one training seed), it reduces AbsRel from 0.011505 to 0.011298 and L1 from 0.013479 to 0.013227 m. It narrowly outperforms a separately trained residual CNN on these two metrics, but not on RMSE, RGB-edge-based boundary error or SSIM. We report these observations with their limitations and identify the controlled additional experiments necessary to evaluate architectural advantage.

1 Introduction

Dense metric depth is central to robotics and scene understanding. RGB alone cannot resolve scale, while mobile range sensors supply low-resolution or sparse depth. Image-guided depth completion combines these signals [3, 5].

Pretrained visual representations, including DINOv2 [4] and Depth Anything [6], enable capable dense depth predictors. PromptDA [2] accepts metric depth prompts and provides accurate dense estimates. Yet downstream images retain localized residual errors. We investigate a small learnable postprocessing module that improves these predictions while keeping PromptDA completely frozen.

Our spatially adaptive consistency gating (SACG) module predicts a residual from RGB, coarse depth and frozen DPT decoder features, then modulates it using a learned gate. The gate uses RGB–depth gradient agreement and a spatial field derived from the validity pattern of the low-resolution LiDAR measurements. The refinement CNN itself does not ingest raw LiDAR depth values.

In the available ARKitScenes seed-10 ablation, the model trained without the optional boundary-weighted loss slightly outperforms both frozen PromptDA and a separately trained simple residual CNN on L1 and AbsRel. We therefore select it as the working method, but do not infer a general architectural advantage from this single-seed comparison.

Contributions of this working manuscript are (1) a code-grounded specification of frozen-PromptDA residual refinement with a consistency-conditioned spatial gate, (2) a transparent comparison to an independently trained plain CNN and component ablations, and (3) a measured report of metric-specific improvements and limitations. Claims about zero-shot generalization, calibrated confidence and state-of-the-art accuracy are out of scope.

2 Related Work

Guided depth completion. Sparse-to-Dense [3] fuses RGB and sparse range measurements. NLSPN [5] applies learned nonlocal spatial propagation to refine depth. Hence spatial refinement itself is not a new invention; our comparison must establish whether the particular implementation adds value relative to simple controls.

Prompt-based depth foundation models. DINOv2 [4] supplies reusable visual features, and Depth Anything [6] scales depth pretraining. PromptDA [2] combines an image with a metric depth prompt. We adopt frozen pretrained PromptDA ViT-L and use its predictions and intermediate DPT features without updating the backbone or decoder.

Gated residual heads. The SACG gate depends on normalized RGB–depth edge agreement and the spatial layout of valid depth observations. It should not be described as calibrated uncertainty. A separately trained plain CNN is an essential baseline; it directly receives resized LiDAR depth and validity, whereas SACG additionally receives frozen DPT features but only a validity-derived LiDAR gating field. This input mismatch limits a pure causal attribution to gating.

3 Method

3.1 Frozen backbone and coarse depth

Let $I \in [0,1]^{3\times H\times W}$ be RGB, S the low-resolution metric LiDAR prompt, and $D^\star$ the reference. A pretrained, frozen PromptDA P_θ produces

$$D_0 = P_\theta(I, S), \qquad \nabla_\theta \mathcal{L} = 0. \tag{1}$$

A forward hook exposes frozen decoder features F from `depth_head.scratch.output_conv1`, all under `torch.no_grad()`.

3.2 Gradient consistency

The code calculates Sobel-gradient magnitudes and normalizes each image by its maximum. Let

$$E(X) = \frac{\sqrt{(K_x * X)^2 + (K_y * X)^2 + \epsilon}}{\max_{u,v} \sqrt{(K_x * X)^2 + (K_y * X)^2 + \epsilon}}. \tag{2}$$

RGB is converted to grayscale using the mean channel value; depth is repeated over three channels before using the same routine. The agreement field is

$$C = \operatorname{clip}\left(1 - |E(I) - E(D_0)|, 0, 1\right). \tag{3}$$

This is a heuristic edge agreement score, not a calibrated prediction confidence.

3.3 Learnable LiDAR validity field

With $M = \mathbf{1}[S > 0]$, a Gaussian-initialized, trainable 31×31 convolution forms $U = K_\psi * M$. The code applies per-image min–max normalization and bilinear resizing:

$$L = \operatorname{Resize}\left(\operatorname{clip}\left(\frac{U - \min U}{\max(\max U - \min U, 10^{-6})}, 0, 1\right)\right). \tag{4}$$

It encodes the locations of valid samples, not their numerical depth. Other code paths use exponential distance to a valid pixel (fixed field) or set the gate field to zero (no field), while PromptDA still consumes the metric depth prompt.

3.4 Gate and residual

The gating CNN takes $[C, L, D_0]$ and has one convolution–batch-normalization–ReLU block with 16 channels and a final single-channel convolution:

$$G = \mathrm{sigmoid}(g_\omega([C, L, D_0])). \tag{5}$$

The residual CNN has three convolution–batch-normalization–ReLU blocks with 32 hidden channels plus a final single-channel convolution. It takes RGB, coarse depth and bilinearly resized frozen DPT features:

$$R = f_\phi([I, D_0, \widetilde{F}]), \qquad \boxed{\widehat{D} = D_0 + G \odot R}. \tag{6}$$

Only ψ, ω, ϕ are optimized.

3.5 Selected no-boundary training objective

For valid pixels $\Omega = \{p : D_p^\star > 0.01\,\mathrm{m},\ D_p^\star \text{ finite}\}$, define

$$\mathcal{L}_{\mathrm{L1}} = \frac{1}{|\Omega|} \sum_{p \in \Omega} |\widehat{D}_p - D_p^\star|, \quad \mathcal{L}_{\mathrm{gate}} = \frac{1}{HW} \sum_p G_p\, \mathrm{stopgrad}(C_p). \tag{7}$$

The selected objective is

$$\mathcal{L}(t) = \mathcal{L}_{\mathrm{L1}} + \mathbf{1}[t \geq 5000]\, 0.1\, \mathcal{L}_{\mathrm{gate}}. \tag{8}$$

The earlier full method also used $0.5\mathcal{L}_{\mathrm{boundary}}$ where

$$\mathcal{L}_{\mathrm{boundary}} = \mathrm{mean}_{p \text{ over image}} \left[\exp(10 E(I)_p) |\widehat{D}_p - D_p^\star| \mathbf{1}[p \in \Omega] \right]. \tag{9}$$

This additional term is disabled in our selected method. Removing a loss component is a model-selection decision rather than a standalone methodological novelty.

4 Experimental Protocol

Data. We use ARKitScenes [1], whose committed evaluation includes 5,599 validation frames, and a locally prepared ScanNet++ [7] export with 2,000 evaluated frames per run and 20,000 training-subset frames. The latter is not the official PromptDA benchmark protocol. RGB and dense target tensors are 756×1008; the prompt retains its native low-resolution grid. Depth is converted from millimeters to meters. Targets are resized by nearest neighbor. Training uses synchronized horizontal flips, while evaluation uses no augmentation.

Optimization. The pretrained PromptDA ViT-L and legacy DPT head remain frozen. The refinement module uses AdamW with learning rate 10^{-4}, weight decay 10^{-4}, batch size two, cosine scheduler and 10,000 optimization steps. The gate penalty begins at step 5,000; the selected variant disables boundary loss. The best validation-L1 checkpoint is selected.

Baselines. Frozen PromptDA requires no downstream training. A separately trained simple residual CNN has six input channels: RGB (three), PromptDA depth (one), resized LiDAR metric values (one), and validity mask (one). Its three hidden convolution blocks have 56 channels; it is optimized with valid-pixel L1. This **does not fine-tune PromptDA**. Ablations vary LiDAR gate fields and optional boundary loss.

Table 1: Committed ARKitScenes validation, seed 10, identical 5,599 frames. Lower is better for all four metrics.

Method	AbsRel	L1 (m)	RMSE (m)	BoundaryAbsRel
Frozen PromptDA	0.011505	0.013479	0.031950	0.020598
Simple residual CNN	0.011321	0.013257	**0.031793**	**0.020335**
SACG with boundary loss	0.011358	0.013305	0.031846	0.020446
SACG without boundary loss	**0.011298**	**0.013227**	0.031803	0.020382

Table 2: ARKitScenes seed-10 SACG configurations; field ablations retain boundary loss.

Variant	AbsRel	L1 (m)	BoundaryAbsRel
Full SACG with boundary	0.011358	0.013305	0.020446
No LiDAR gate field	0.011396	0.013351	0.020497
Fixed LiDAR gate field	0.011420	0.013385	0.020480
No boundary loss	**0.011298**	**0.013227**	**0.020382**

Metrics and validity. L1, AbsRel and RMSE use finite ground-truth depth strictly above 0.01 m, computed per frame and then averaged across images. RGB-edge BoundaryAbsRel uses Sobel RGB edges (threshold 0.1; dilation 5), not annotated depth discontinuities. SSIM is computed with invalid regions zero-filled and is auxiliary. All seed-10 ARKitScenes rows use the same 5,599-frame manifest (SHA-256 `5b8a182fd4fba0dead6371b54d15c96974d7f5f853074b210b7c50ea3f3dd2b9`). The selected model was chosen using validation results, not an unseen test split; cross-seed ScanNet++ sample sets also vary.

5 Results

5.1 ARKitScenes: selection of the main variant

The selected no-boundary variant improves AbsRel by 1.79% and L1 by 1.87% over frozen PromptDA, with small additional improvements over the simple CNN. However, the CNN has lower RMSE and RGB-edge-based boundary error. Thus these are single-seed, metric-specific gains, not evidence of overall superiority.

5.2 Component ablations

The selected loss changes only the boundary term. The field ablations are instead comparisons under the earlier full loss and therefore do not isolate the field's necessity under the selected objective.

5.3 Existing multi-seed results belong to the earlier full objective

Stored full-objective models improve AbsRel and L1 against frozen PromptDA on each evaluated subset. The ScanNet++ subset varies by seed, and no committed cross-dataset result was found for selected no-boundary SACG. No new checkpoint validation, paired confidence intervals or held-out tests were executed for this draft.

6 Discussion and Conclusion

Removing the exponential RGB-edge boundary loss coincides with lower AbsRel and L1 for one ARKitScenes validation seed. A potential explanation is that the previous loss overweights

Table 3: ARKitScenes results for SACG *with boundary loss*, not the selected model.

Seed	AbsRel	L1 (m)	RMSE (m)
10	0.011358	0.013305	0.031846
26	0.011399	0.013365	0.031891
42	0.011376	0.013315	0.031876

Table 4: ScanNet++ local-export results for full SACG *with boundary loss*, 2,000 frames per seed; not the selected model.

Seed	Baseline AbsRel	Full SACG AbsRel	Baseline L1	Full SACG L1
10	0.029730	0.029613	0.048934	0.048479
26	0.028579	0.028510	0.048643	0.048400
42	0.029356	0.029263	0.048766	0.048388

appearance texture, but this has not been tested causally. SACG without boundary loss still uses edge consistency inside its gate.

The simple CNN is a meaningful and competitive control. It narrowly trails SACG in L1 and AbsRel but improves RMSE, RGB-edge BoundaryAbsRel and SSIM relative to the selected method. Because the CNN consumes metric LiDAR values directly whereas SACG refinement uses frozen DPT features, this is not a clean isolated test of gating.

Limitations include no calibrated uncertainty supervision, no explicit preservation of valid LiDAR observations in refined depth, no guarantee of positive output depth, no independent scene-held-out test, no three-seed results for the selected objective, and no selected-model ScanNet++ evaluation. The evaluation split also influenced method choice.

In summary, SACG combines frozen PromptDA with a learned validity-guided spatial gate and a DPT-feature residual. The no-boundary training variant shows promising but small single-seed improvements. Robust claims about architectural advantage require additional controlled ablations and independent evaluation.

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